Chapter 38 — Shared Shopping Cart Across Screens

> **Part V — Shared & Global State** · Complexity: **Complex** (≥40 interview questions)
>
> **Prerequisites:** Ch 3 (SSOT), Ch 6 (UI state/UDF), Ch 29 (offline-first), Ch 31 (outbox), Ch 33 (conflicts), Ch 11 (single-flight), Ch 15 (idempotency), Ch 42 (multi-account, previewed). This chapter is the **canonical shared-state problem** that ties Parts I–IV together into one feature.
>
> **Reference surface in StreamCart:** the shopping cart — edited from the product list, product detail, a mini-cart badge, and the cart screen; working offline; merging guest→user on login; consistent across devices.

38.1 Problem Definition and Real-World Context

A shopping cart is the textbook **shared, mutable, cross-screen state**: the *same* cart is read and written from many screens (catalog, product detail, mini-cart badge, cart screen, checkout), must stay **consistent everywhere instantly**, work **offline**, survive process death, **merge** a guest cart into the user's cart on login, and sync **across devices**. It's the single best vehicle for shared-state architecture, and the naive approaches fail spectacularly:

- **Per-screen cart copies.** Each screen fetches/holds its own cart → the badge says 2 while the cart screen shows 3; adding on detail doesn't update the list. There's no single source of truth.
- **Cart in a singleton with raw mutable state.** A `CartManager` with a mutable list and manual listeners leaks, races under concurrent edits, and loses state on process death.
- **Blocking/online-only cart.** "Add to cart" hangs or fails offline; users lose items.
- **Lost guest cart.** A guest adds items, logs in, and their cart is empty (not merged) — a conversion killer.
- **Cross-device divergence.** Cart edited on phone and web diverge with no conflict policy.

The architecture applies the book's patterns to one feature: **one cart SSOT** (Room, Ch 3) observed by every screen; **optimistic writes** (Ch 6/29) via an **outbox** (Ch 31) with **idempotency** (Ch 15); **single-flight** sync (Ch 11); **guest→user merge** on login; **conflict resolution** (Ch 33) for concurrent edits; and **correct scoping/lifecycle** (shared ViewModel vs SSOT, Ch 40/42). This chapter shows how the pieces compose into a production cart.

Three practical use cases

1. **E-commerce (StreamCart Shop).** The running example — add/remove/change quantity from any screen; badge + cart + checkout always agree; offline + sync.
2. **OTT (Watch).** "My list" / download queue — the same shared-collection pattern (add from browse, detail, player).
3. **Food/travel (generic).** A booking cart / order basket edited across steps, persisted, merged on login, synced across devices.

Why this is architectural

The cart crystallizes **shared-state design**: single source of truth, cross-screen consistency, state ownership/scoping, offline + optimistic writes, merge semantics, and conflict handling — the decisions that determine whether shared state stays coherent. It's a favorite interview because it exercises the whole stack.

38.2 Requirements Engineering

Functional requirements

ID	Requirement
F1	One cart, read/written from any screen; all screens reflect changes instantly.
F2	Add / remove / change quantity; compute totals (subtotal, count, price).
F3	Work offline: writes succeed locally immediately; sync later.
F4	Survive process death / rotation.
F5	Merge a guest cart into the user's cart on login.
F6	Sync across devices with conflict resolution for concurrent edits.
F7	Handle item availability/price changes from the server.

Non-functional requirements

Attribute	Policy
Consistency	Single source of truth; every screen coherent; eventual cross-device consistency.
Availability	Fully usable offline (Ch 29).
Reliability	No lost cart edits (outbox, Ch 31); idempotent (Ch 15).
Performance	Instant local reads/writes; no refetch per screen.
Correctness	Concurrent edits resolved (Ch 33); accurate totals.
Security	Per-account cart; cleared on logout (Ch 22/42).

Scope

- **In scope:** cart SSOT + cross-screen observation, optimistic writes + outbox + idempotency, offline, process-death survival, guest→user merge, cross-device conflict resolution, state ownership/scoping, totals.
- **Out of scope:** checkout/payment flow (Ch 75), wishlist (Ch 74), the underlying patterns' internals (Ch 3/29/31/33 — reused), multi-account mechanics (Ch 42).

Assumptions & constraints

- Room (Ch 3), optimistic writes + outbox (Ch 29/31), idempotency (Ch 15), conflict resolution (Ch 33). Backend has a cart API + merge endpoint. `minSdk 24`.

38.3 System Design Fundamentals

- **Single Source of Truth for the cart.** The cart lives in **Room** (Ch 3), exposed by **one `CartRepository`** (a DI singleton, Ch 5). **Every screen's ViewModel observes the same repository `Flow`** — so one write updates the badge, the list, the cart screen, and checkout simultaneously. This is the core fix for cross-screen consistency: *don't share a ViewModel or a mutable object — share the SSOT.*
- **State ownership, not a god-singleton.** The cart's mutable state is owned by the SSOT (Room via the repository), not a hand-rolled mutable singleton. Screens are stateless observers (UDF, Ch 6); they emit events (`add`, `remove`), the repository mutates the SSOT, and all observers re-render.
- **Optimistic writes (Ch 6/29).** `addItem` updates Room immediately (UI reflects it at once) and enqueues an outbox mutation (Ch 31) with an idempotency key (Ch 15); the server write happens asynchronously. Offline works by construction.
- **Totals as derived state.** Subtotal/count/price are **computed from** the cart items (a mapping/`combine`), never stored and kept in sync manually — derive, don't duplicate.
- **Guest→user merge.** A guest has a local (device-scoped) cart; on login, **merge** the guest cart with the server/user cart (union items; sum or max quantities per policy) so nothing is lost. This is a one-time conflict resolution (Ch 33) between two carts.
- **Cross-device concurrency.** The same user's cart edited on two devices diverges; resolve with a policy — often **sum deltas** (a CRDT-ish counter per line) or LWW per line, so both devices' intent is honored (Ch 33).
- **Scoping (Ch 40/42).** The cart is **per account** (cleared on logout, Ch 22); within the app it's an app-scoped singleton (not Activity/ViewModel-scoped) so it survives navigation. (Ch 40 covers shared ViewModel scoping as an *alternative* for narrower sharing.)

Core insight: **the cart is shared state solved by a single Room-backed SSOT that every screen observes, mutated optimistically through an outbox with idempotency, derived totals, guest→user merge, and conflict resolution across devices** — not a shared mutable object. Share the *source of truth*, not the *state holder*.

38.4 High-Level Architecture


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Components & ownership

- **`CartRepository`** (singleton) owns the SSOT; the only writer; every screen reads it.
- **Room `cart_items`** (per account) is the SSOT, observed by all screens' ViewModels.
- **Outbox + idempotency** (Ch 31/15) make writes durable and exactly-once.
- **Sync + conflict** (Ch 29/30/33) reconcile with the server/other devices.
- **Merge** folds a guest cart into the user cart on login.

Direction: any screen → repository → Room SSOT (all screens observe) + outbox → sync ↔ server.

38.5 Detailed Component Design

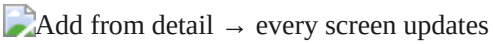


Component	Responsibility	Note
`CartRepository` (singleton)	SSOT owner; single writer; optimistic + outbox	Ch 3/29/31
Room `cart_items`	durable per-account cart	SSOT; observed by all screens
derived `Cart` (totals)	computed from lines	derive, don't store
per-screen `CartViewModel`	observe + emit events	UDF (Ch 6); may be shared (Ch 40)
merge	guest→user on login	conflict (Ch 33)

> **Every screen gets its own ViewModel that injects the *same* `CartRepository`** — not a shared ViewModel holding the cart. The SSOT is shared, the ViewModels aren't. (Ch 40 discusses when a shared ViewModel is appropriate instead.)

38.6 End-to-End Data Flow

Add from detail → every screen updates



Guest → user merge on login



38.7 Complete Kotlin Implementation

Package: `com.streamcart.data.cart` and `:feature:cart`.

38.7.1 Cart entity + SSOT DAO (per account)

```
@Entity(tableName = "cart_items", indices = [Index(value = ["accountId","productId"], unique =
true)])
data class CartItemEntity(
    @PrimaryKey val id: String,                // client-generated (works offline)
    val accountId: String,                     // per-account scoping (Ch 42)
    val productId: String,
    val quantity: Int,
    val unitPriceCents: Long,
    val updatedAt: Long,
    val version: Long = 0,                      // conflict detection (Ch 33)
    val syncStatus: SyncStatus = SyncStatus.PENDING,
    val deleted: Boolean = false,               // soft delete (syncs)
)

@Dao
interface CartDao {
    @Query("SELECT * FROM cart_items WHERE accountId = :acct AND deleted = 0 ORDER BY
updatedAt")
    fun observe(acct: String): Flow<List<CartItemEntity>>   // every screen observes this
    @Upsert suspend fun upsert(item: CartItemEntity)
    @Query("SELECT * FROM cart_items WHERE accountId = :acct AND productId = :pid AND deleted =
0")
    suspend fun findByProduct(acct: String, pid: String): CartItemEntity?
    @Query("DELETE FROM cart_items WHERE accountId = :acct") suspend fun clearAccount(acct:
String)
}
```

38.7.2 Cart repository: single SSOT owner, optimistic + outbox

```
@Singleton
class CartRepositoryImpl @Inject constructor(
    private val db: StreamCartDatabase,
    private val dao: CartDao,
    private val outbox: OutboxDao,                // Ch 31
    private val account: AccountProvider,         // current accountId (Ch 42)
    private val syncScheduler: SyncScheduler,     // Ch 30
    private val clock: Clock, private val json: Json,
) : CartRepository {

    // SSOT -> derived Cart (totals computed, not stored).
    override fun observeCart(): Flow<Cart> =
        account.currentId().flatMapLatest { acct -> dao.observe(acct) }
            .map { lines -> Cart(
                lines = lines.map { it.toDomain() },
                subtotal = Money(lines.sumOf { it.unitPriceCents * it.quantity }),
                itemCount = lines.sumOf { it.quantity },
            ) }

    override suspend fun add(productId: String, qty: Int) {
        val acct = account.currentIdValue()
        db.withTransaction {                                    // atomic optimistic + outbox
(Ch 31)
            val existing = dao.findByProduct(acct, productId)
            val line = existing?.copy(quantity = existing.quantity + qty, updatedAt =
clock.now(),
                                      version = existing.version + 1, syncStatus =
SyncStatus.PENDING)
                ?: CartItemEntity(UUID.randomUUID().toString(), acct, productId, qty,
lookupPrice(productId), clock.now())
            dao.upsert(line)
            outbox.insert(mutation("cart.upsert", line.id,
json.encodeToString(line.toRequest())))
        }
        syncScheduler.requestSync()                             // Ch 30
    }

    override suspend fun setQuantity(lineId: String, qty: Int) {
        if (qty <= 0) return remove(lineId)
        mutateLine(lineId) { it.copy(quantity = qty) }
    }
    override suspend fun remove(lineId: String) = mutateLine(lineId) { it.copy(deleted = true) }
// soft delete
```

```
    private suspend fun mutateLine(lineId: String, f: (CartItemEntity) -> CartItemEntity) {
        db.withTransaction {
            val cur = dao.findById(lineId) ?: return@withTransaction
            val next = f(cur).copy(updatedAt = clock.now(), version = cur.version + 1,
syncStatus = SyncStatus.PENDING)
            dao.upsert(next)
            outbox.insert(mutation("cart.upsert", lineId,
json.encodeToString(next.toRequest())))
        }
        syncScheduler.requestSync()
    }

    // Guest -> user merge on login (Ch 33): union lines; sum quantities per policy.
    override suspend fun mergeGuestCart(guestAccountId: String) {
        val user = account.currentIdValue()
        db.withTransaction {
            dao.observe(guestAccountId).first().forEach { g ->
                val existing = dao.findByProduct(user, g.productId)
                val mergedQty = (existing?.quantity ?: 0) + g.quantity        // sum policy
                dao.upsert((existing ?: g).copy(id = existing?.id ?:
UUID.randomUUID().toString(),
                    accountId = user, quantity = mergedQty, syncStatus = SyncStatus.PENDING,
version = (existing?.version ?: 0) + 1))
                outbox.insert(mutation("cart.upsert", existing?.id ?: g.id, /* payload */ ""))
            }
            dao.clearAccount(guestAccountId)                                  // drop guest cart
        }
        syncScheduler.requestSync()
    }
}
```

38.7.3 Per-screen ViewModels share the repository (not a shared VM)

```
// Badge, detail, cart screens each inject the SAME singleton repository.
@HiltViewModel
class MiniCartViewModel @Inject constructor(repo: CartRepository) : ViewModel() {
    val count: StateFlow<Int> = repo.observeCart().map { it.itemCount }
        .stateIn(viewModelScope, SharingStarted.WhileSubscribed(5_000), 0)
}

@HiltViewModel
class CartViewModel @Inject constructor(private val repo: CartRepository) : ViewModel() {
    val uiState: StateFlow<CartUiState> = repo.observeCart()
        .map { if (it.lines.isEmpty()) CartUiState.Empty else CartUiState.Content(it) }
        .stateIn(viewModelScope, SharingStarted.WhileSubscribed(5_000), CartUiState.Loading)
    fun onEvent(e: CartUiEvent) = viewModelScope.launch {
        when (e) {
            is CartUiEvent.ChangeQty -> repo.setQuantity(e.lineId, e.qty)     // optimistic
            is CartUiEvent.Remove -> repo.remove(e.lineId)
            is CartUiEvent.Add -> repo.add(e.productId, 1)
        }
    }
}
```

38.7.4 Conflict-aware sync apply (cross-device, Ch 33)

```
// When the server returns an updated cart line (delta, Ch 32), resolve vs local pending.
suspend fun applyServerLine(remote: CartItemEntity, resolver: ConflictResolver<CartItemEntity>)
{
    val local = dao.findById(remote.id)
    when (val r = resolver.resolve(base = dao.findBase(remote.id), local = local, remote =
remote)) {
        is Resolution.Accept -> dao.upsert(r.remote)
        is Resolution.Merged -> { dao.upsert(r.value); outbox.insert(/* push merged */) }
        else -> { /* keep local / user-resolved */ }
    }
}
```

38.7.5 Module structure

```
:data:cart     CartRepositoryImpl (SSOT, optimistic, outbox, merge)
:core:database CartItemEntity, CartDao (per-account)
:feature:cart  CartViewModel, MiniCartViewModel, DetailAddToCart (share the repo)
:sync          outbox drain (Ch 31), conflict resolver (Ch 33)
```

38.8 Concurrency and Race Conditions

Scenario	Risk	Handling
Add from two screens fast	duplicate lines / lost update	unique (accountId,productId) index; upsert-increment; version
Optimistic write + sync in flight	overwrite	version + outbox (Ch 31); server dedups (Ch 15)
Two devices edit same line	divergence	conflict resolution (Ch 33): sum-deltas / LWW per line
Guest→user merge race	lost items	merge in a transaction; idempotent push
Process death mid-edit	lost edit	durable SSOT + outbox (Ch 31)
Badge vs cart screen inconsistency	stale UI	both observe the same SSOT `Flow` — can't diverge
Account switch	wrong cart	per-account scoping; `flatMapLatest` on accountId (Ch 42)

- **One SSOT observed by all screens makes cross-screen consistency automatic** — the badge and cart can't disagree because they read the same `Flow`. This is the whole point of shared *source of truth* over shared *state holder*.
- **Unique (account, product) + upsert-increment** prevents duplicate lines when the same product is added from two places concurrently.

38.9 Failure Scenarios and Recovery

Failure	Detection	Response	UX
Offline add	connectivity (Ch 68)	optimistic local + outbox	works; "will sync"
Sync rejected (out of stock)	server error	mark line unavailable; surface; keep others	"X no longer available"
Price changed on server	delta (Ch 32)	update unit price; recompute totals; notify	"price updated"
Concurrent device edit	version conflict	resolve (Ch 33: sum/LWW)	merged
Guest cart lost on login	merge missing	implement guest→user merge	items preserved
Process death	durable SSOT+outbox	restored	seamless
Item removed on server	tombstone (Ch 32)	remove locally	cart updated
Account switch	per-account	load that account's cart; clear on logout (Ch 22)	correct cart



> **Availability + accuracy:** the cart must stay usable offline (optimistic) **and** accurate (reconcile price/stock/conflicts on sync) — surface server-driven changes (out of stock, price) rather than silently diverging at checkout.

38.10 Scaling the Implementation

- **Level 1.** In-memory cart (single screen) — toy; loses everything.
- **Level 2.** Room SSOT + one repository observed by all screens (cross-screen consistency, survives rotation/process death).
- **Level 3 — production.** Optimistic writes + outbox (Ch 31) + idempotency (Ch 15)
    - offline (Ch 29); per-account scoping; derived totals; server reconcile (price/ stock).
- **Level 4 — large.** Cross-device sync (Ch 32) + conflict resolution (Ch 33); guest→user merge; promotions/coupons; real-time cart updates (WebSocket, Ch 63).
- **Level 5 — enterprise.** Server-authoritative cart with client replica, inventory reservation, distributed consistency, A/B-tested merge/coupon logic, cart analytics/abandonment.

Avoid overengineering: a simple app may not need cross-device sync or conflict resolution — Room SSOT + optimistic + outbox covers most. Add sync/merge when guest carts and multi-device genuinely matter (most commerce apps do).

38.11 Alternative Approaches and Trade-offs

Dimension	Option A	Option B	When
Shared state	**Shared SSOT** (Room repo, all screens observe)	shared ViewModel / singleton with mutable state	SSOT always for cross-screen durable state; shared VM only for narrow, ephemeral sharing (Ch 40)
Writes	Optimistic + outbox	pessimistic (block on server)	Optimistic for UX/offline; pessimistic only if server confirmation is mandatory pre-UI
Totals	Derived (computed)	stored + maintained	Derive — storing invites drift
Authority	Client replica + server sync	server-only (fetch every change)	Replica+sync for offline/speed; server-only loses offline
Cross-device	Conflict resolution (Ch 33: sum/LWW)	last-write-wins blind	Policy per semantics; blind LWW loses intent
Guest cart	Merge on login	discard	Merge — discarding loses conversions
Scope	App-scoped singleton repo	Activity/ViewModel-scoped	App-scoped so it survives navigation; narrower scope loses cross-screen sharing

Shared SSOT vs shared ViewModel (the core decision): the classic "how do screens share the cart" question. A **shared ViewModel** (Ch 40) scoped to an Activity/nav graph can share state between a few screens, but it's **ephemeral** (lost on process death unless SavedState), tied to a navigation scope, and doesn't survive the app or sync across devices. The **SSOT** (Room repository) is durable, app-wide, offline, and the single writer — the right choice for the cart. **Share the source of truth, not the state holder.** Use a shared ViewModel only for transient UI state shared by a tight screen group (Ch 40).

Optimistic vs pessimistic cart writes: optimistic (update UI now, sync later) gives instant feedback and offline support but must reconcile (stock/price/conflicts) on sync; pessimistic (wait for server) is simpler/accurate but unusable offline and laggy. Carts are optimistic, with explicit reconcile at sync/checkout.

38.12 Security Considerations

- **Per-account cart; clear on logout** (Ch 22/42) — one user's cart must never show for another; scope by `accountId` and wipe on logout.
- **Server authorizes and re-prices** every cart mutation and the final checkout — a tampered client could set quantity/price locally, but the **backend is authoritative** on price, availability, and limits (never trust client-computed totals for payment, Ch 75).
- **Idempotency** (Ch 15) on cart mutations prevents duplicate lines from retries.
- **Guest cart** is device-scoped, not backed up with secrets (Ch 37); merge over an authenticated channel (Ch 19).
- **Encrypt** the cart DB if it contains sensitive data (usually not, but PII in custom fields → Ch 24).
- Client vs backend: the client holds a convenient replica for UX; the backend owns the authoritative cart, pricing, and inventory — the client's totals are display- only until server-confirmed at checkout.

38.13 Performance and Resource Optimization

- **Instant local reads/writes** (SSOT) — no network on add/view; excellent perceived performance.
- **One `Flow` observed by all screens** — no per-screen refetch; `WhileSubscribed` stops idle collection.
- **Derived totals** computed cheaply from lines (O(n)); memoize if large.
- **Batch/debounce rapid quantity changes** (stepper spam) before enqueuing outbox (Ch 31 compaction) to cut sync volume.
- **Single-flight** cart fetch (Ch 11) to avoid stampede when multiple screens open.
- **Profiling:** cross-screen update latency, cart DB size, sync volume, recomposition of badge/cart.

38.14 Testing Strategy

In-memory Room + fakes; assert cross-screen consistency, optimistic+outbox, merge, conflicts.

```
class CartRepositoryTest {
    @Test fun `add reflects across all observers (one SSOT)`() = runTest {
        val repo = repo()
        val badge = repo.observeCart().map { it.itemCount }
        val cart = repo.observeCart()
        repo.add("p1", 2)
        assertEquals(2, badge.first())
        assertEquals(1, cart.first().lines.size)          // same SSOT -> consistent
    }

    @Test fun `add works offline and queues outbox`() = runTest {
        val repo = repo(online = false)
        repo.add("p1", 1)
        assertEquals(1, repo.observeCart().first().itemCount)   // optimistic
        assertEquals(1, outbox.pending().size)                  // queued (Ch 31)
    }

    @Test fun `adding same product twice increments, no duplicate line`() = runTest {
        val repo = repo(); repo.add("p1", 1); repo.add("p1", 2)
        val cart = repo.observeCart().first()
        assertEquals(1, cart.lines.size); assertEquals(3, cart.itemCount)
    }

    @Test fun `totals are derived correctly`() = runTest {
        val repo = repo(prices = mapOf("p1" to 1000))
        repo.add("p1", 3)
        assertEquals(3000, repo.observeCart().first().subtotal.cents)
    }

    @Test fun `guest cart merges into user cart on login`() = runTest {
        val repo = repo(account = "guest"); repo.add("p1", 2)
        account.switchTo("user"); repo.mergeGuestCart("guest")
        assertEquals(2, repo.observeCart().first().itemCount)   // preserved
    }

    @Test fun `concurrent device edit resolved by policy`() = runTest {
        // local qty 2 (v1), remote qty 3 (v2) -> sum or LWW per resolver (Ch 33)
    }

    @Test fun `cart cleared on logout`() = runTest {
        val repo = repo(); repo.add("p1", 1); session.logout()
        assertTrue(repo.observeCart().first().lines.isEmpty())
    }
}
```

Also: process-death survival (SSOT durable), price/stock reconcile on sync, idempotent double-sync, per-account isolation, a Compose test that the badge and cart screen update together.

38.15 Production Observability

- **Metrics:** cart conversion, add-to-cart rate, cart-sync success/failure, offline- cart usage, guest→user merge success, cross-device conflict rate, cart-abandonment, cross-screen inconsistency incidents (should be 0), price/stock reconcile rate.
- **Logs:** `cart{op, line_count, result}`, `cart_merge{guest_items, result}`, `cart_conflict{resolution}` (ids/counts, no PII, Ch 28), correlation IDs (Ch 1/84).
- **Debugging:** badge ≠ cart screen → not sharing the SSOT (per-screen copies); lost items offline → non-durable/no outbox; empty cart after login → merge missing; duplicate lines → missing unique index / no upsert-increment; wrong cart after switch → not per-account scoped.
- **Alerting:** cart-sync failure or merge-failure spike → backend/sync incident.
- **Privacy:** cart is user data; redact; clear on logout.

38.16 System Design Interview Preparation

> 40+ questions. (A canonical "design the shared cart" interview.)

Fundamentals

1. *How do multiple screens share the cart?* One SSOT (Room repository); every screen's ViewModel observes the same `Flow` — not a shared ViewModel/mutable singleton.
2. *Why not a shared ViewModel holding the cart?* Ephemeral (lost on process death), nav-scoped, no offline/sync; share the *source of truth*, not the holder.
3. *Why Room SSOT?* Durable, offline, survives process death, single writer, observable by all.
4. *How do the badge and cart stay consistent?* They read the same DAO `Flow` → one write updates both.
5. *How are totals kept in sync?* Derived (computed) from lines, never stored.

Writes / offline

6. *Add to cart offline?* Optimistic local write + outbox (Ch 31) + idempotency (Ch 15); syncs later.
7. *Why idempotency?* Retries/replays shouldn't duplicate lines (Ch 15).
8. *Client-generated line ids?* So adds work offline before the server assigns ids.
9. *Rapid quantity changes?* Debounce/compact before sync (Ch 31).

Consistency / concurrency

10. *Add same product from two screens?* Unique (account,product) + upsert-increment → one line.
11. *Two devices edit the cart?* Conflict resolution (Ch 33): sum-deltas (CRDT-ish) or LWW per line.
12. *Process death mid-edit?* Durable SSOT + outbox → restored.
13. *Optimistic vs pessimistic?* Optimistic for UX/offline; reconcile on sync.

Guest / account

14. *Guest→user merge?* On login, union lines, sum/LWW quantities, push merged (Ch 33).
15. *Why merge (not discard)?* Don't lose the guest's cart (conversion).
16. *Per-account?* Scope by accountId; clear on logout (Ch 22/42).
17. *Account switch?* `flatMapLatest` on accountId → load that cart.

Server reconcile

18. *Price changed?* Delta (Ch 32) updates unit price; recompute totals; notify.
19. *Out of stock?* Mark line unavailable at sync/checkout; surface.
20. *Who owns price/availability?* The backend (authoritative); client totals are display-only (Ch 75).

Scoping/lifecycle

21. *Repository scope?* App-scoped singleton (survives navigation), per-account data.
22. *When a shared ViewModel instead?* Narrow, ephemeral sharing of transient UI state among a tight screen group (Ch 40).
23. *Survive rotation?* ViewModel + SSOT; `WhileSubscribed`.

Security

24. *Trust client totals for payment?* No — backend re-prices at checkout (Ch 75).
25. *Cross-account leak?* Per-account scope + clear on logout.
26. *Idempotency for payment-adjacent ops?* Yes (Ch 15).

Performance

27. *Avoid per-screen refetch?* One shared SSOT `Flow`.
28. *Stepper spam?* Compact outbox (Ch 31).
29. *Cart fetch stampede on open?* Single-flight (Ch 11).

Testing

30. *How test cross-screen consistency?* Multiple observers of the repo; one write updates all.
31. *How test offline add?* Fake offline; assert optimistic + outbox.
32. *How test merge?* Guest cart + login → assert preserved.

Production debugging

33. *Badge ≠ cart?* Not sharing the SSOT.
34. *Items lost offline?* No durable outbox.
35. *Empty after login?* No merge.
36. *Duplicate lines?* No unique index/upsert.
37. *Wrong cart after switch?* Not per-account.

Design/scale

38. *Real-time cart (price drops)?* WebSocket push (Ch 63) → update SSOT.
39. *Inventory reservation?* Server-authoritative; client shows pending.
40. *Cross-device conflict policy?* Sum-deltas CRDT or LWW per line (Ch 33).

Design-change challenges

1. **Cart must work offline.** Optimistic + outbox + idempotency (Ch 29/31/15).
2. **Guest adds then logs in.** Guest→user merge (Ch 33).
3. **Edited on phone + web.** Per-line conflict resolution (Ch 33).
4. **Prices change server-side.** Delta reconcile (Ch 32) + surface.
5. **Badge and cart disagree.** Single SSOT observed by both.

Mock walkthrough (abridged)

Clarify (cross-screen? offline? guest? multi-device?) → one Room SSOT + repository; all screens observe → optimistic writes + outbox + idempotency → derived totals → per-account scope + clear on logout → guest→user merge + cross-device conflict (Ch 33) → server reconcile price/stock → "share SSOT not state holder" → testing consistency.

38.17 Practical Exercises

1. **Beginner.** Back the cart with one Room SSOT + repository; show a badge and cart screen that both update from a single add. *Eval:* consistent across screens; survives rotation.
2. **Intermediate.** Add optimistic writes + outbox + idempotency; prove offline add works and syncs. *Eval:* instant UI; queued; no duplicate on double-sync.
3. **Advanced.** Add guest→user merge on login and per-account scoping with clear-on-logout. *Eval:* guest cart preserved; no cross-account leak.
4. **Production debugging.** Fix a cart where the badge and cart screen disagree (per-screen copies) and items vanish offline (no outbox). *Eval:* one SSOT; durable.
5. **System design challenge.** Add cross-device sync with per-line conflict resolution (sum/LWW) and server price reconcile. *Eval:* converges; prices updated; intent preserved.

(Solutions withheld unless requested.)

38.18 Chapter Summary & Reference Sheet

Key decisions: solve shared cross-screen state with **one Room-backed SSOT** (a singleton `CartRepository`) that **every screen's ViewModel observes — share the source of truth, not the state holder**; mutate **optimistically** through an **outbox** (Ch 31) with **idempotency** (Ch 15) so it works **offline** and survives process death; **derive totals** (don't store); scope the cart **per account** and **clear on logout** (Ch 22/42); **merge guest→user** on login and resolve **cross-device conflicts** per line (sum/LWW, Ch 33); **reconcile** server price/ stock on sync; the **backend is authoritative** on price/availability (never trust client totals for payment, Ch 75).

Concepts: shared SSOT vs shared ViewModel, cross-screen consistency via one `Flow`, optimistic writes + outbox + idempotency, derived state, per-account scoping, guest→user merge, cross-device conflict resolution, server reconcile.

Common mistakes: per-screen cart copies (badge ≠ cart); shared mutable singleton (races/leaks/no durability); online-only/blocking writes (no offline); storing totals (drift); no guest merge (lost cart); no per-account scope (cross-user leak); duplicate lines (no unique index/upsert); trusting client totals for payment.

Checklist:

- [ ] one Room SSOT + singleton repo
- [ ] all screens observe same `Flow`
- [ ] optimistic + outbox + idempotency (Ch 31/15)
- [ ] offline (Ch 29)
- [ ] derived totals
- [ ] unique (account,product) + upsert-increment
- [ ] per-account + clear on logout
- [ ] guest→user merge (Ch 33)
- [ ] cross-device conflict (Ch 33)
- [ ] server reconcile price/stock
- [ ] backend authoritative at checkout.

Key APIs/patterns: Room SSOT + DAO `Flow` (Ch 3), `@Singleton` repository (Ch 5), optimistic write + `withTransaction` + outbox (Ch 31), idempotency keys (Ch 15), `flatMapLatest` on account (Ch 42), `combine` / `map` derived totals, `ConflictResolver` (Ch 33), `SharingStarted.WhileSubscribed`.

References:

- Guide to app architecture (SSOT/shared state): https://developer.android.com/topic/architecture
- Data layer / offline-first: https://developer.android.com/topic/architecture/data-layer/offline-first
- ViewModel scopes (shared VM alternative, Ch 40): https://developer.android.com/topic/libraries/architecture/viewmodel
- (composes) Ch 3 SSOT, Ch 6 UDF, Ch 29 Offline, Ch 31 Outbox, Ch 33 Conflicts, Ch 42 Multi-Account

Chapter review (Phase 4)

- **Completeness:** all 18 sections; architecture/class/sequence/failure diagrams; cart entity + SSOT repo (optimistic/outbox/merge) + per-screen VMs + conflict apply; concurrency (one-SSOT consistency, unique index); failure (offline/stock/price/merge/ conflict); 5 levels; SSOT-vs-shared-VM and optimistic-vs-pessimistic trade-offs; security (per-account, server authoritative); perf (one Flow, derived totals); cross-screen + merge tests; observability (badge≠cart); 40 Q&A + 5 challenges. ✅
- **Expand later:** checkout/payment (Ch 75), shared ViewModel scoping (Ch 40), multi-account (Ch 42), real-time cart (Ch 63).
- **Interview coverage:** exceeds ≥40 bar. **Opens Part V.**